COMP9602: Optimization

Convex Programs (II)

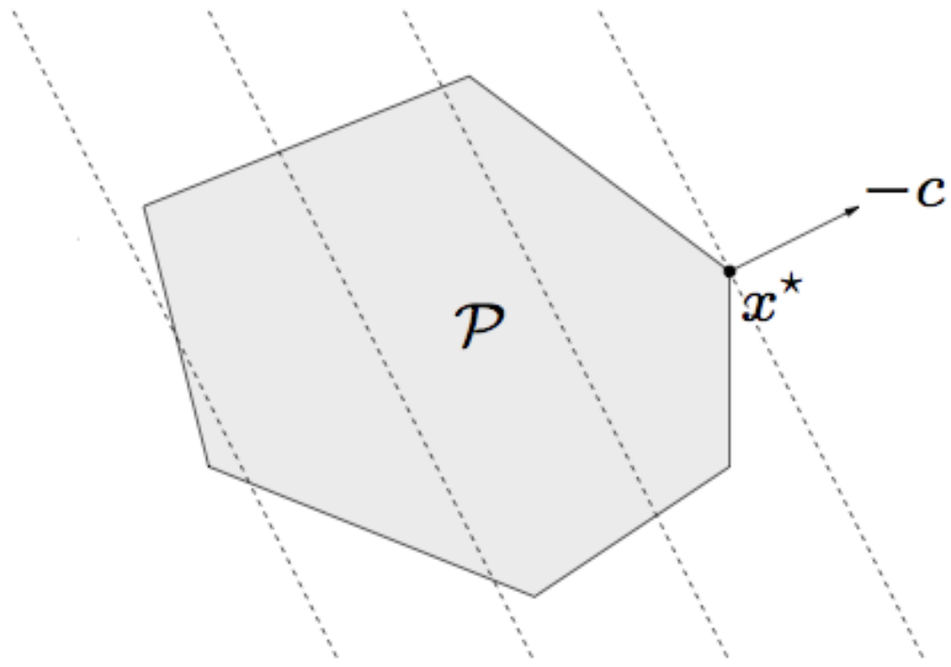
Prof. Chuan Wu

Department of Computer Science
The University of Hong Kong

Linear program (LP)

$$\begin{array}{ll} \text{minimize} & c^T x + d \\ \text{subject to} & Gx \preceq h \\ & Ax = b \end{array}$$

- convex problem with affine objective and constraint functions
- feasible set is a polyhedron



Standard form LP

- Standard form

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b, \quad x \succeq 0 \end{array}$$

- Canonical form

$$\begin{array}{l} \text{minimize } c^T x \\ \text{subject to } Ax \succeq b, x \succeq 0 \end{array}$$

convertible to standard form

Examples

- The diet problem
- Network flow problems



Examples

- Piecewise-linear minimization

$$\text{minimize} \quad \max_{i=1,\ldots,m}(a_i^T x + b_i)$$

equivalent to an LP

$$\begin{array}{ll} \text{minimize} & t \\ \text{subject to} & a_i^T x + b_i \leq t, \quad i = 1, \ldots, m \end{array}$$

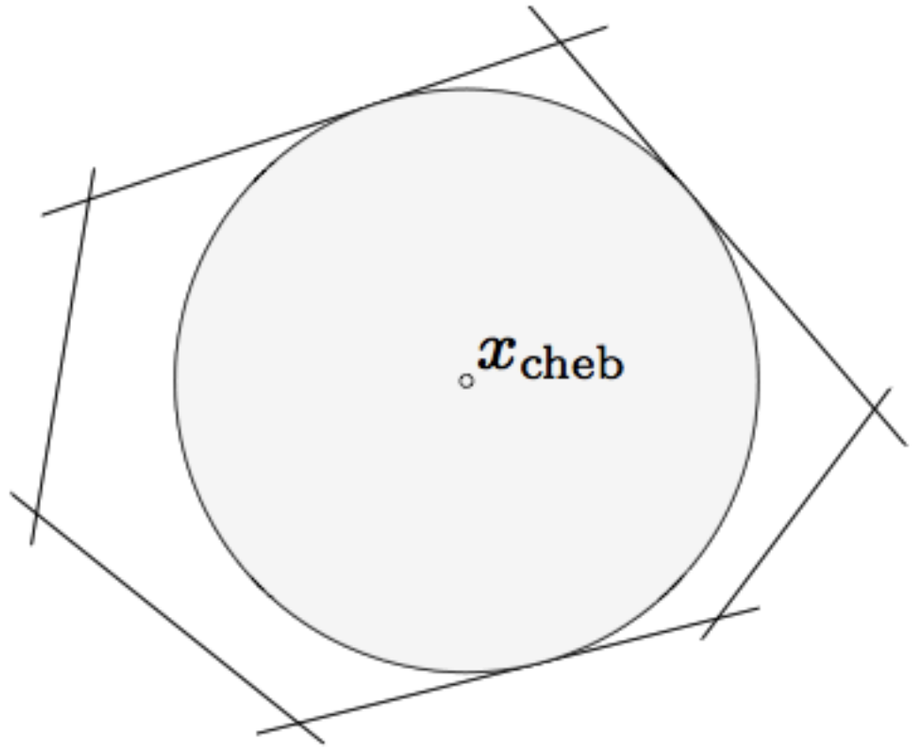
Examples

- Largest sphere inside a polyhedron

$$\mathcal{P} = \{x \mid a_i^T x \leq b_i,\ \ i = 1, \ldots, m\}$$

  - Chebyshev center of a polyhedron: center of the largest sphere

$$\mathcal{B} = \{x_c + u \mid \|u\|_2 \leq r\}$$



- hence, x_c, r can be determined by solving the LP

$$\begin{array}{ll} \text{maximize} & r \\ \text{subject to} & a_i^T x_c + r\|a_i\|_2 \leq b_i, \quad i = 1, \ldots, m \end{array}$$

Examples

- Derivation of the LP

$$\max \quad r$$
$$\text{s.t. } x_c + u \in P, \quad \forall ||u||_2 \leq r$$

$$\Updownarrow$$

$$a_i^T(x_c + u) \leq b_i, i = 1, \ldots, m, \forall ||u||_2 \leq r$$

$$\Updownarrow$$

$$a_i^T x_c + a_i^T u \leq b_i, i = 1, \ldots, m, \forall ||u||_2 \leq r$$

$$\Updownarrow$$

$$a_i^T x_c + \sup_{||u||_2 \leq r} a_i^T u \leq b_i, i = 1, \ldots, m$$

$$\Updownarrow$$

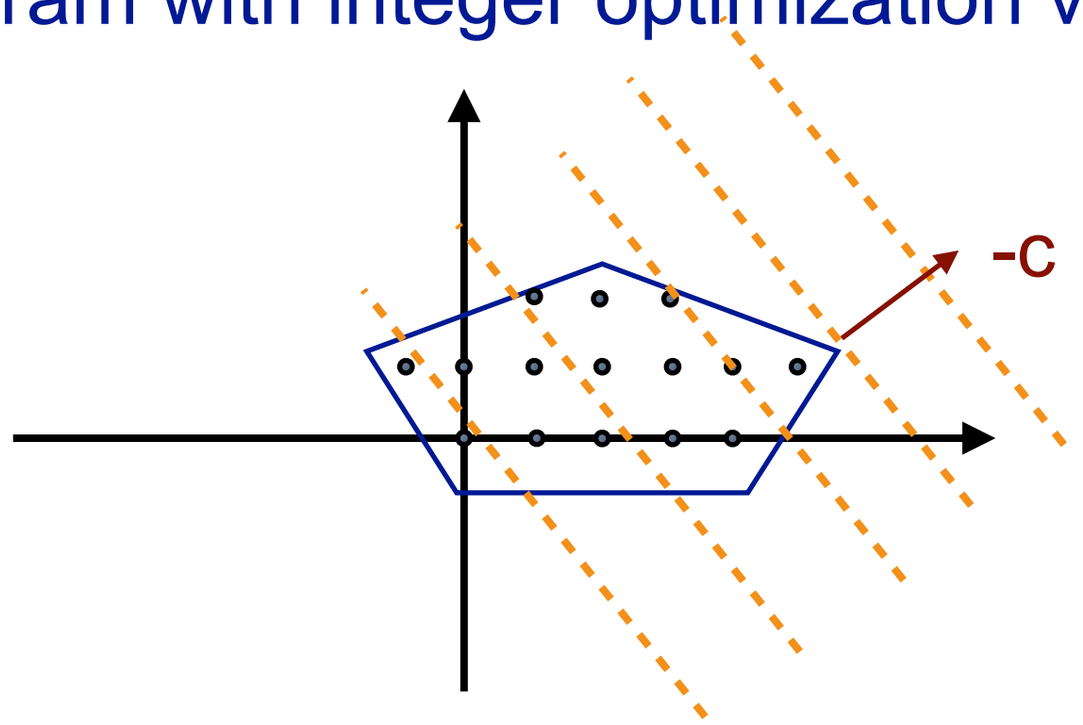
$a_i^T u$ is largest when $u = \dfrac{a_i}{||a_i||_2} r$

$$a_i^T x_c + a_i^T \frac{a_i}{||a_i||_2} r \leq b_i, i = 1, \ldots, m$$

Integer (linear) program (IP, ILP)

$$\begin{array}{ll} \text{minimize} & c^T x + d \\ \text{subject to} & Gx \preceq h \\ & Ax = b \\ & x \in \mathbf{Z}^n \end{array}$$

- Linear program with integer optimization variables



- Application of linear program to combinatorial optimization problems

Example

- Shortest path problem
  - find a path from s to t in directed graph G=(V,E) with the smallest total cost

$$\min \sum_{(i,j)\in E} c_{ij}x_{ij}$$

subject to:

$$\sum_{j:(i,j)\in E} x_{ij} - \sum_{j:(j,i)\in E} x_{ji} = \begin{cases} 1, \text{for } i = s, \\ 0, \text{for all } i \in V - \{s,t\}, \\ -1, \text{for } i = t, \end{cases}$$

$$x_{ij} \in \{0,1\}, \forall (i,j) \in E$$

Example

- Minimum vertex cover
  - find the smallest subset of vertices in undirected graph G=(V,E) that contains at least one endpoint of every edge in the graph

$$\min \sum_{i \in V} x_i$$

subject to:

$$x_i + x_j \geq 1, \forall (i,j) \in E$$

$$x_i \in \{0,1\}, \forall i \in V$$

Example

- Maximum independent set
  - find the largest subset of vertices in undirected graph G=(V,E) such that no two vertices in the subset are connected by an edge in the graph

$$\max \sum_{i \in V} x_i$$

subject to:

$$x_i + x_j \leq 1, \forall (i,j) \in E$$

$$x_i \in \{0,1\}, \forall i \in V$$

Example

- Maximum weighted bipartite matching
  - find the subset of edges in a bipartite graph, that has the largest total weight, such that each vertex is incident to exactly 1 edge in the subset (perfect matching)

$$\max \sum_{(i,j)\in E} w_{ij} x_{ij}$$

subject to:

$$\sum_{j:(i,j)\in E} x_{ij} = 1, \forall i \in V$$

$$x_{ij} \in \{0,1\}, \forall (i,j) \in E$$

Example

- Maximum weight b-matching
  - find the subset of the edges in an undirected graph that has the largest total weight, such that each vertex is incident to at most b edges in the subset

$$\max \sum_{e \in E} w_e x_e$$

subject to:

$$\sum_{\text{e incident on v}} x_e \leq b, \forall v \in V$$

$$x_e \in \{0,1\}, \forall e \in E$$

NP hardness of integer programming

- Integer programming problems are generally NP-hard
  - ILPs that are NP hard

    minimum vertex cover, maximum independent set

    (they are complementary problems)
  - ILPs that are not NP hard

    shortest path problem, maximum weight bipartite matching, b matching

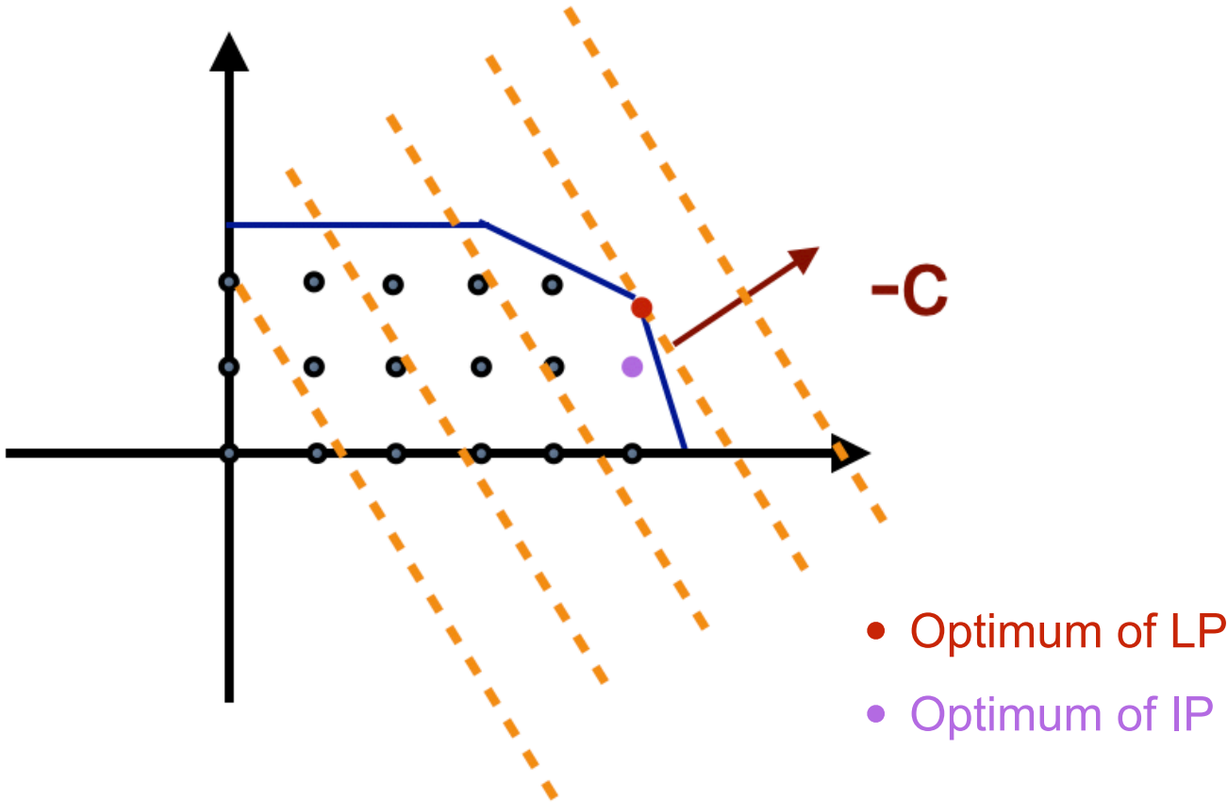
Solving integer programs

- One natural idea for solving ILPs: "relax" the integrality constraint, i.e., allow x to take on real values

  the resulting problem is an LP

  problem: the LP's optimum can be different from the optimum of the ILP

integrality gap: $\sup_I \frac{OPT(I)}{OPT_f(I)}$



OPT(I): exact optimal value of the ILP

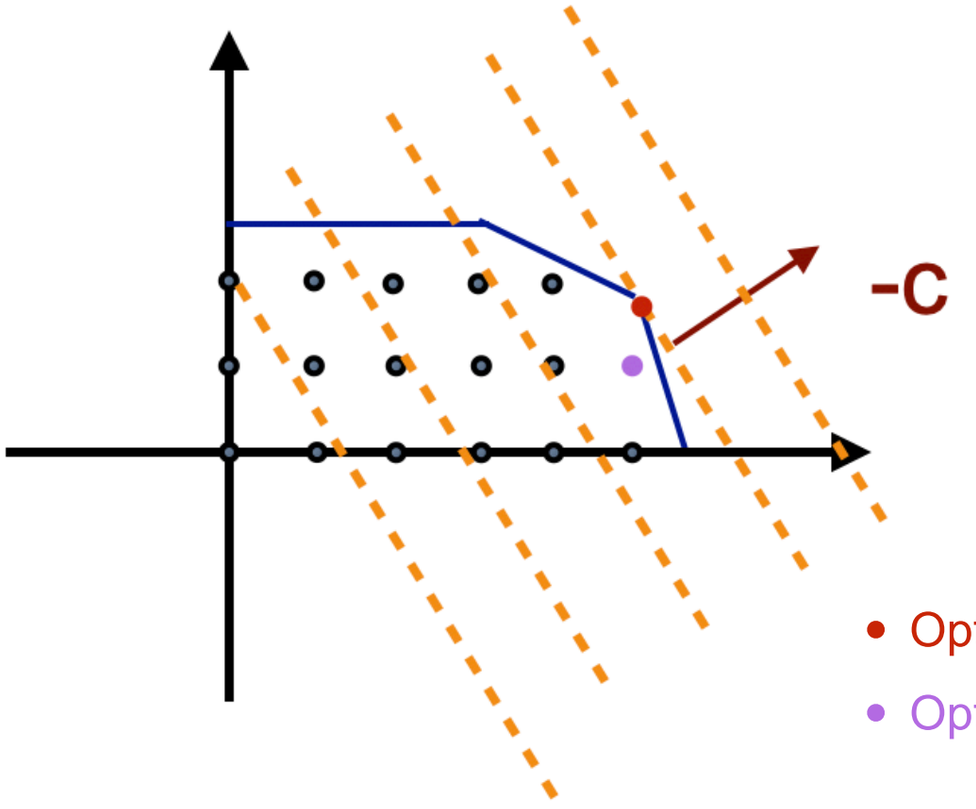
$OPT_f(I)$: optimal value of the relaxed LP

the max. ratio between OPT(I) and $OPT_f(I)$, over all instances (I) of the problem

Solving integer programs (cont'd)

solution: try to round the solution of the LP to a feasible integral point and take that as the solution to the ILP

- determining how to do rounding to achieve the optimum can be as hard as solving the original ILP itself
- develop approximation algorithms that go from the optimal point of the LP to a point that is nearby the optimal point of the ILP, and bound how far off it is from the true optimal point



approximation ratio:

$$\alpha = \sup_I \frac{A(I)}{OPT(I)}$$

A(I): approximate optimal value

OPT(I): exact optimal value of the integer program

Solving integer programs (cont'd)

Example: solving the minimum vertex cover problem by relaxing the integer constraints to $x_i \geq 0, \forall i \in V$, and rounding the solution of the LP to obtain the (approximate) solution to the ILP

Let x_i^* be the optimal solution of the LP, we have $x_i^* \in \{0, \frac{1}{2}, 1\}$

Then $\sum_{i \in V} x_i \leq \sum_{i \in V} 2x_i^* = 2OPT_f(I) \leq 2OPT(I)$

Approximation ratio: $\alpha = 2$

Total unimodularity

There are special cases that the relaxation is exact,
i.e., the relaxed LP provides an integer optimal point

- Total unimodular matrix

  An integer matrix is totally unimodular (TUM) if the determinant of every square submatrix of the matrix is either -1, 0, or 1.

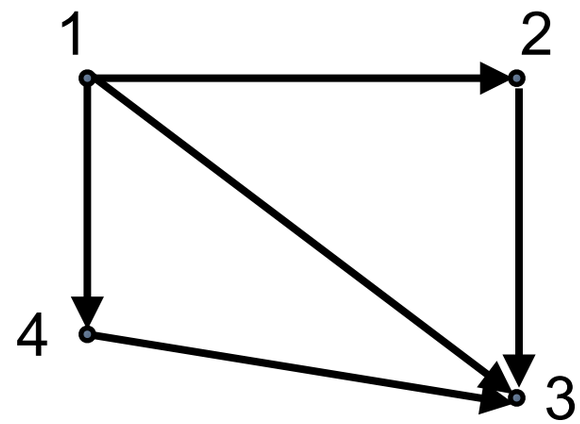
  e.g., incidence matrix of a directed graph;
  incidence matrix of an undirected bipartite graph

Total unimodularity (cont'd)

Examples

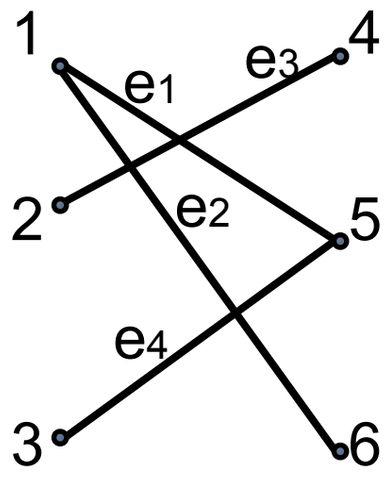
- Incidence matrix of a directed graph

Vertex \ Edge	(1,2)	(1,3)	(1,4)	(2,3)	(4,3)
1	-1	-1	-1		
2	1			-1	
3		1		1	1
4			1		-1



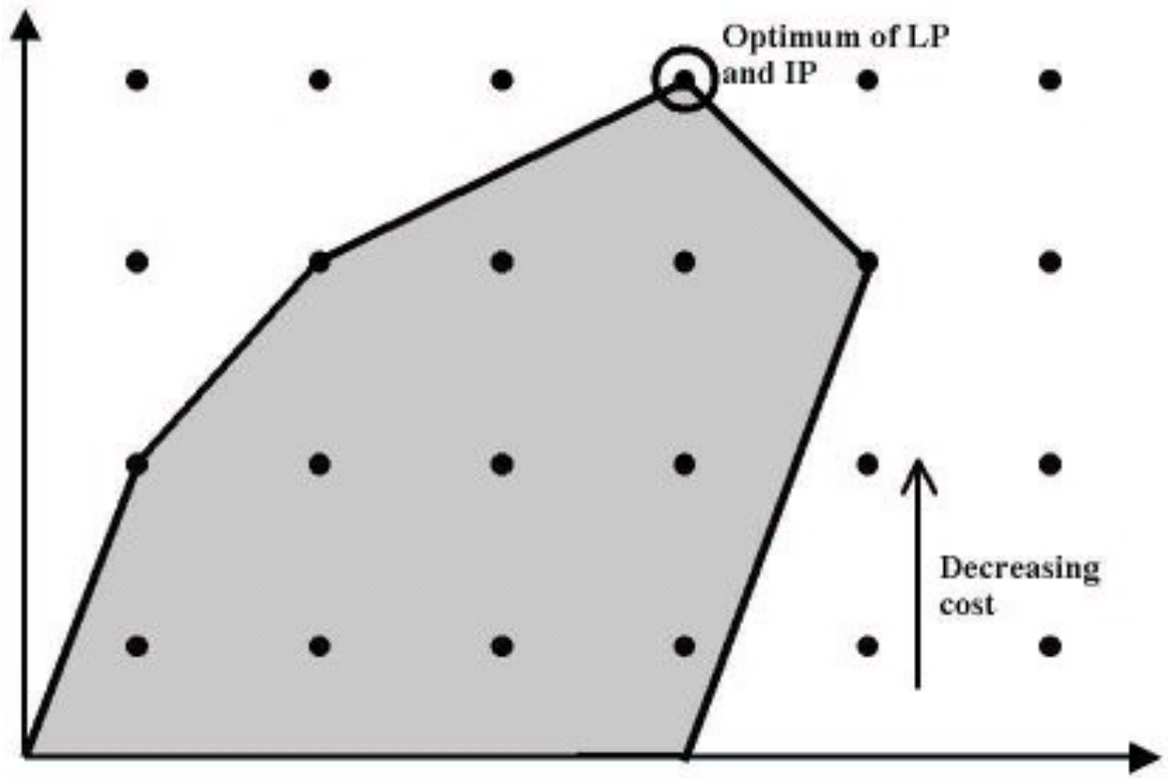
- Incidence matrix of an undirected graph

Vertex \ Edge	e_1	e_2	e_3	e_4
1	1	1		
2			1	
3				1
4			1	
5	1			1
6		1		



Total unimodularity (cont'd)

- If A is TUM and b is an integer vector, all vertices of the polyhedron $P = \{x | Ax = b, x \succeq 0\}$ (or $P = \{x | Ax \succeq b, x \succeq 0\}$) are integer

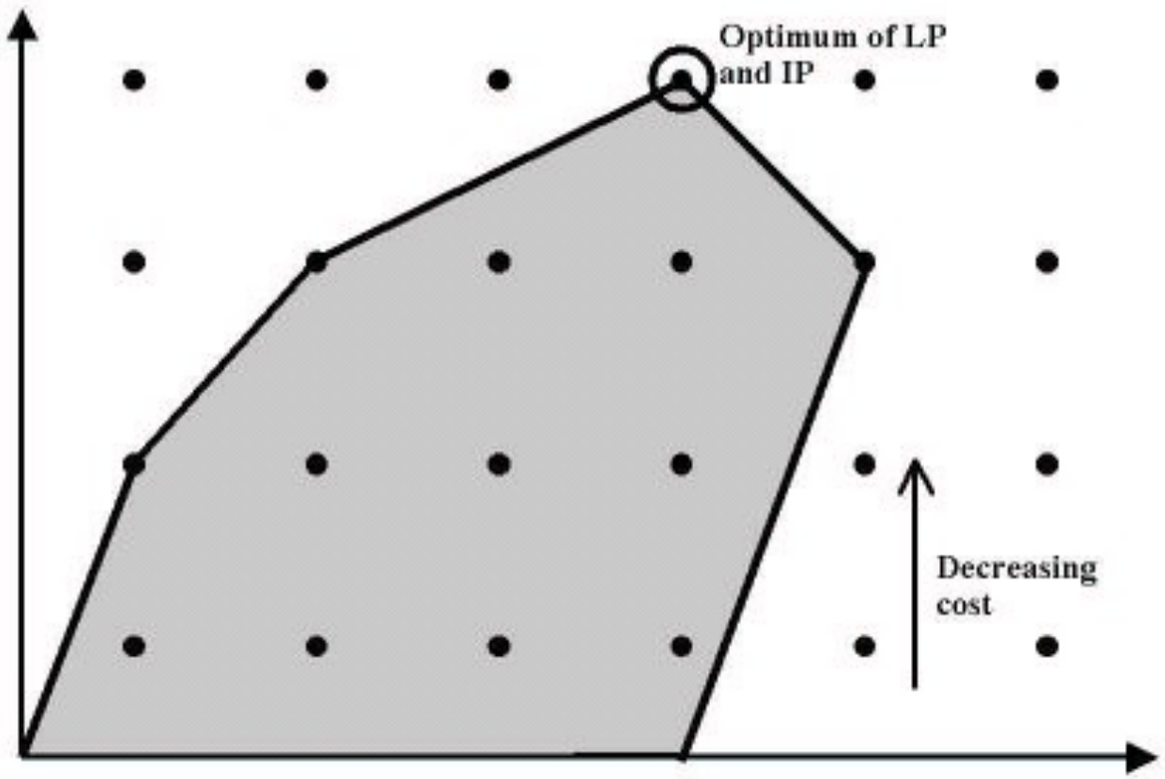


Total unimodularity (cont'd)

- An LP in the standard or canonical form where constraint matrix A is TUM and b is an integer vector, has an integer optimal point

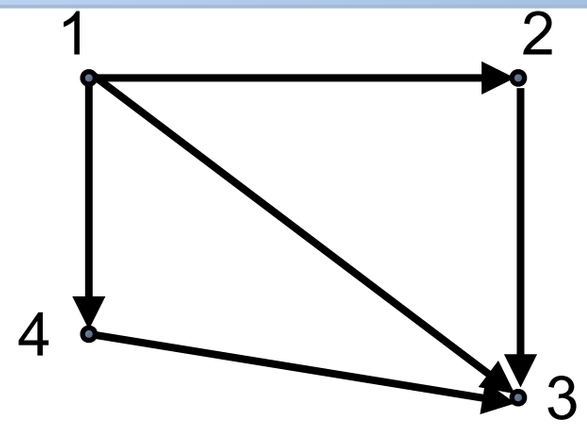
  So an ILP whose LP relaxation satisfies the above conditions can be solved exactly by solving its LP relaxation!

  e.g., LP relaxation of shortest path problem, weighted bipartite matching



Total unimodularity (cont'd)

Example: LP relaxation of shortest path problem



$$\min \sum_{(i,j)\in E} c_{ij}x_{ij}$$

subject to:

$$\sum_{j:(i,j)\in E} x_{ij} - \sum_{j:(j,i)\in E} x_{ji} = \begin{cases} 1, \text{for } i = s, \\ 0, \text{for all } i \in V - \{s,t\}, \\ -1, \text{for } i = t, \end{cases}$$

$$x_{ij} \in \{0,1\}, \forall (i,j) \in E$$

$$x_{12} + x_{13} + x_{14} = 1$$
$$-x_{12} + x_{23} = 0$$
$$\Rightarrow \quad -x_{13} - x_{23} - x_{43} = -1$$
$$-x_{14} + x_{43} = 0$$

$$x \succeq 0$$

$$A = \begin{bmatrix} -1 & -1 & -1 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & -1 \end{bmatrix} \qquad b = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Reference

- Chapter 4.3, Convex Optimization.
- For integer program: Chapter 13.1, Combinatorial Optimization: Algorithms and Complexity

Acknowledgement

- Some materials are extracted from the slides of Stanford EE364a and EE364b
- Some materials are extracted from the lecture notes of Convex Optimization by Prof. Wei Yu at the University of Toronto